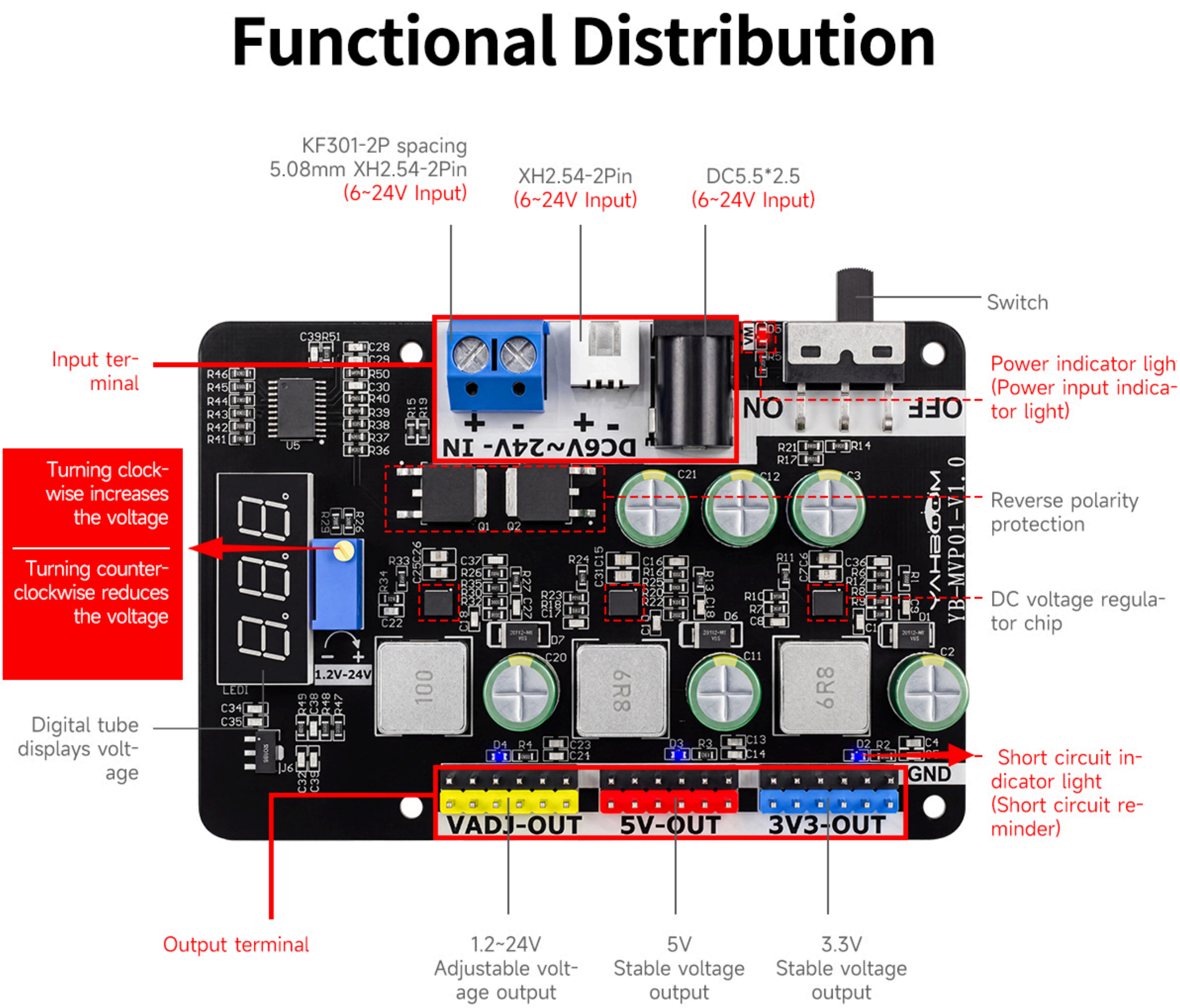
# How to Use the Adjustable Step-Down Voltage Regulator Module



## 1. Functional Schematic

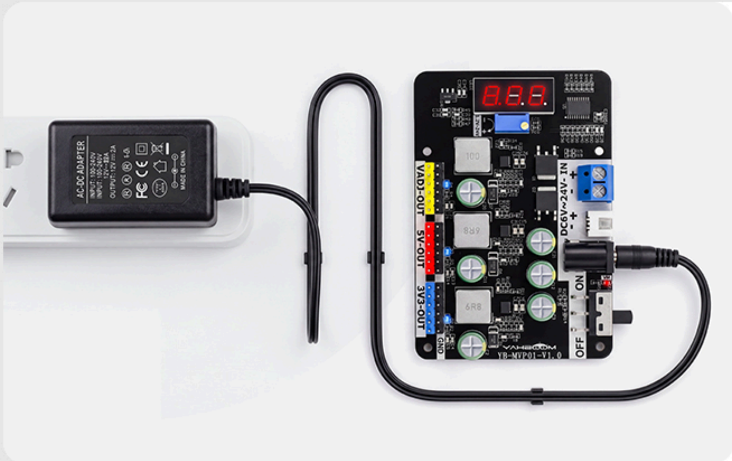


## 2. Power Input

Connect the power supply to the input terminals; the acceptable voltage range is 6V to 24V. Flip the switch to the left position to turn on the module; the red indicator LED will light up, indicating that the board is receiving power normally. If the indicator does not light up, check the following: ensure the power supply is not connected in reverse polarity, verify that the power connection is secure (no loose contacts), and confirm that the switch is indeed flipped to the left. Connecting the power supply in reverse will *not* damage the module, as it features built-in reverse-polarity protection; simply disconnect the power supply and reconnect it with the correct polarity to resume operation.

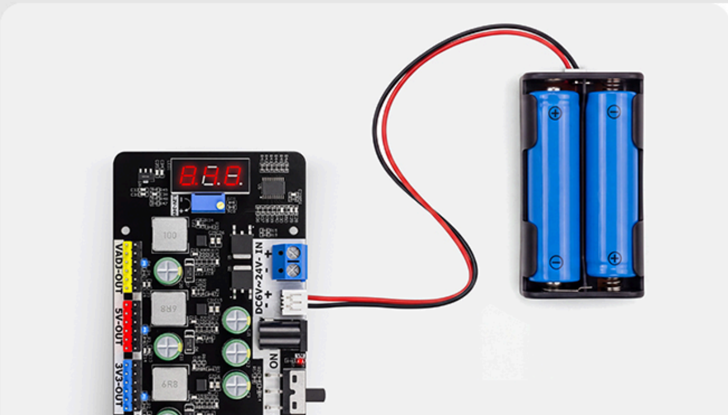
The following diagram illustrates various methods for powering the module, allowing users to choose the option that best suits their needs.

# Support Multiple Power Inputs



**01**

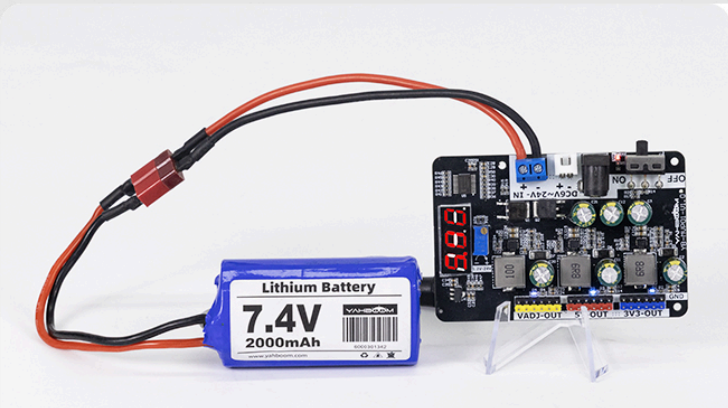
DC5.5*2.5 interface con-
nected to power adapter



*With XH2.54 power line

**02**

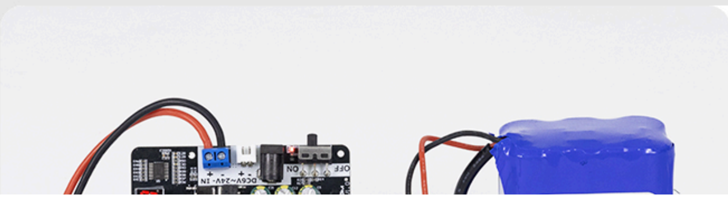
XH2.54 interface connect-
ed to battery box input




*Need to purchase a battery that includes a T-type power line

**03**

Connect the wiring termi-
nal to the 7.4V battery
pack input

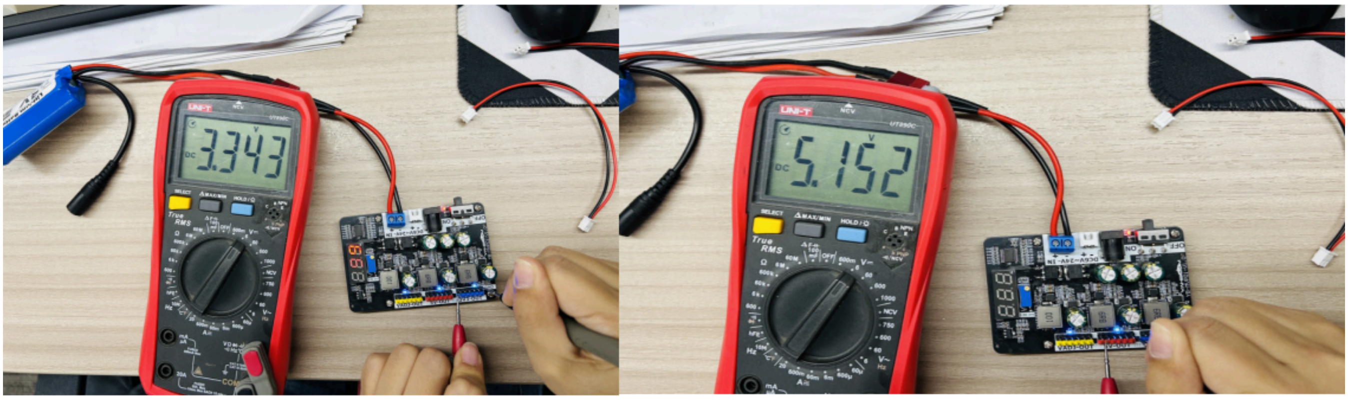


**04**

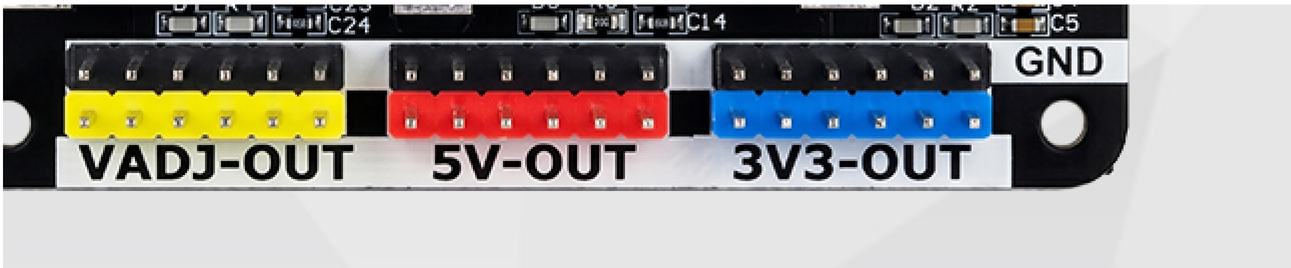

## 3. Load Output

The output terminals provide three distinct voltage outputs: 3.3V, 5V, and an adjustable voltage. The adjustable voltage range spans from 1.2V to 24V.

The image below displays the actual measured output voltages for the 5V and 3.3V terminals.



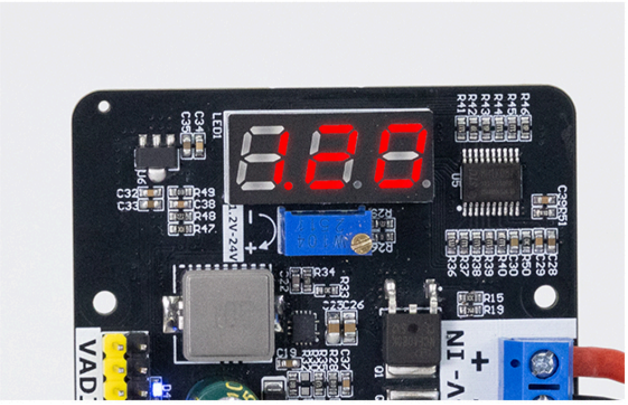

The black pin header serves as the Ground (GND) terminal; the blue pin header outputs 3.3V; the red pin header outputs 5V; and the yellow pin header outputs the adjustable voltage. Refer to the diagram below for the pin layout.



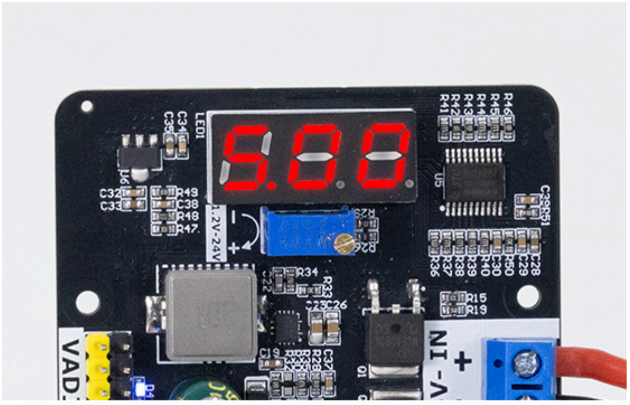

The adjustable voltage output range is 1.2V to 24V. The images below illustrate how the digital display appears at various voltage settings.

# Adjustable Output Voltage With Digital Display

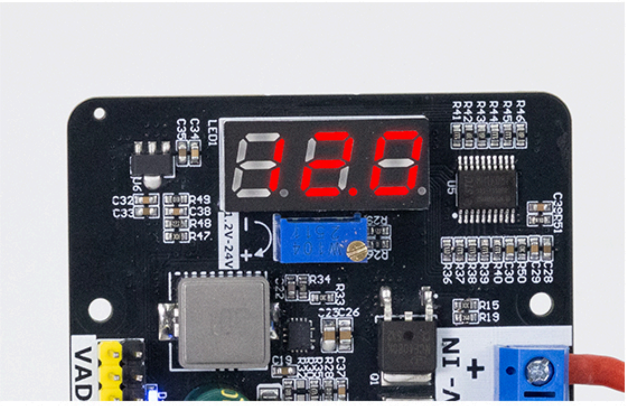
The voltmeter shows an error of ±1%



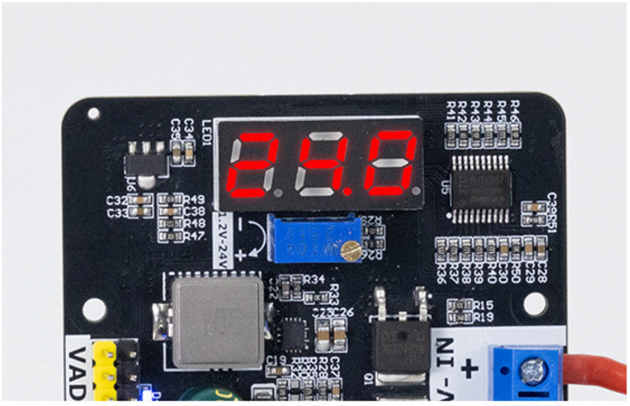

1.2V Accurate and adjustable




5V Standard voltage regulation



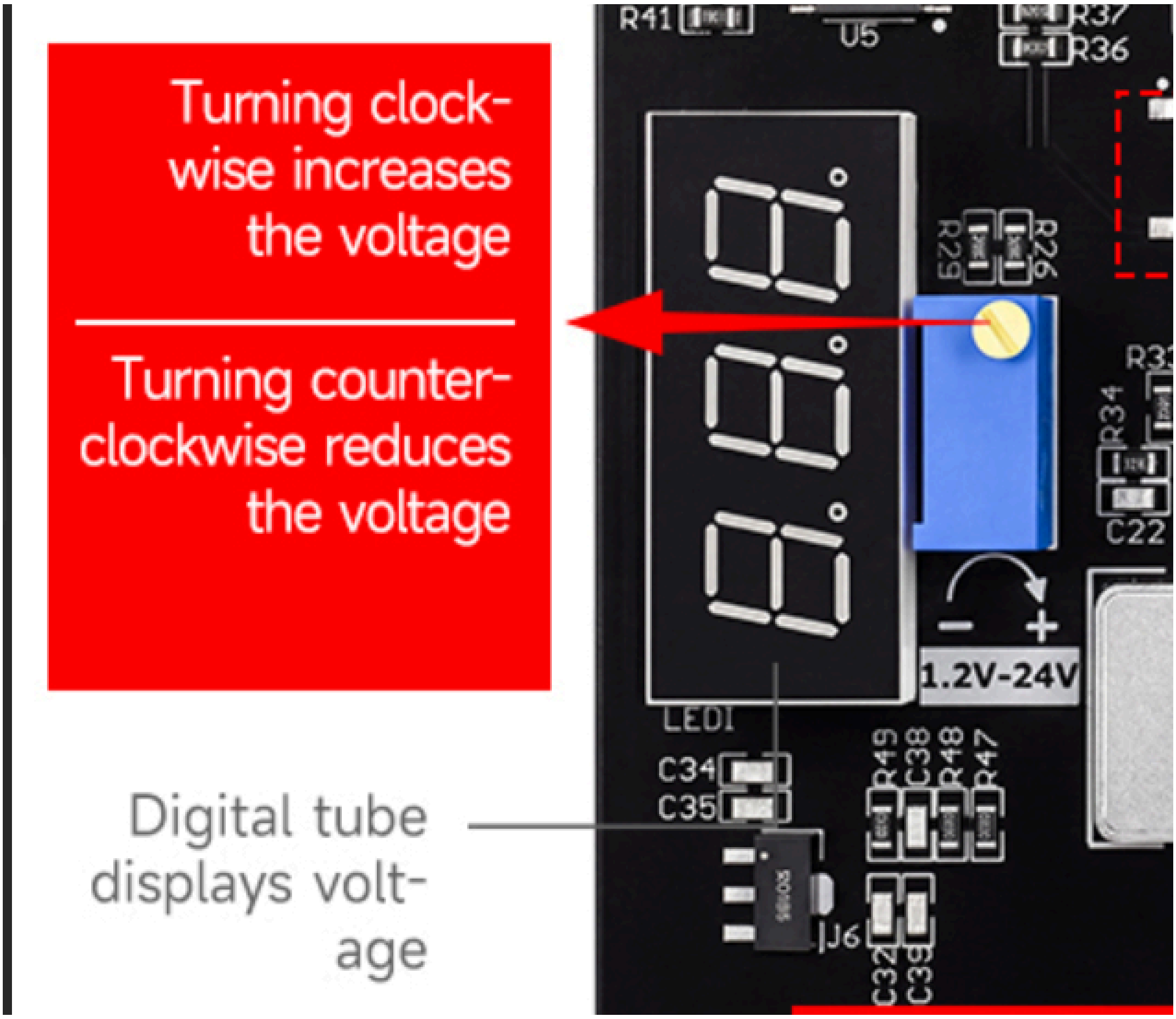

12V Power supply does not jump voltage




24V Full power output

Using the Adjustable Potentiometer:

Use the flat-head screwdriver included in the accessory kit to adjust the output voltage. Turning the potentiometer clockwise (↻) increases the voltage, while turning it counter-clockwise (↺) decreases the voltage. When the potentiometer reaches its maximum or minimum limit, it will emit a faint "click" sound; this indicates that it has reached its mechanical limit and should not be turned any further to prevent damage to the potentiometer. The digital display shows only the value of the adjustable output voltage, with an accuracy tolerance of ±1%.

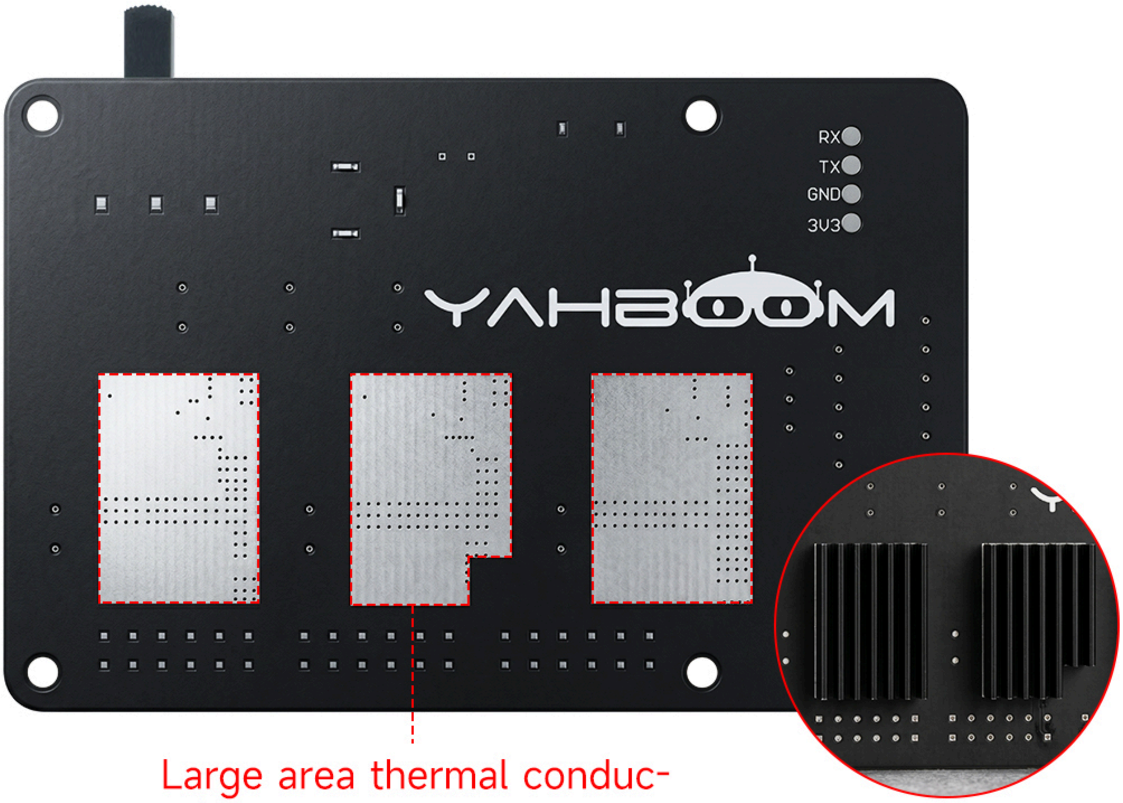



Note: If the input voltage is below 20V, it is possible for the output voltage to reach its maximum value (24V) *before* the adjustment knob reaches its mechanical limit. In this scenario, the knob may still be turned further even though the output voltage will not increase; this is considered normal behavior. The adjustable potentiometer has a resistance value of 100K; the knob can only be rotated to its full extent when the input voltage reaches 24V.

This module is a step-down (buck) converter; therefore, the input voltage must be higher than the adjustable output voltage. There must be a minimum voltage difference of 0.5V between the input and output, a requirement dictated by the DC power supply chip used in the module. The maximum current for a single output channel is 5A. When operating at 5A, proper heat dissipation is critical; if conditions permit, it is recommended to attach a heatsink. During operation, place the product on a non-conductive flat surface to prevent accidental contact with metal objects, which could lead to a short circuit.

# Large area thermal conductive zone can enhance heat dissipation

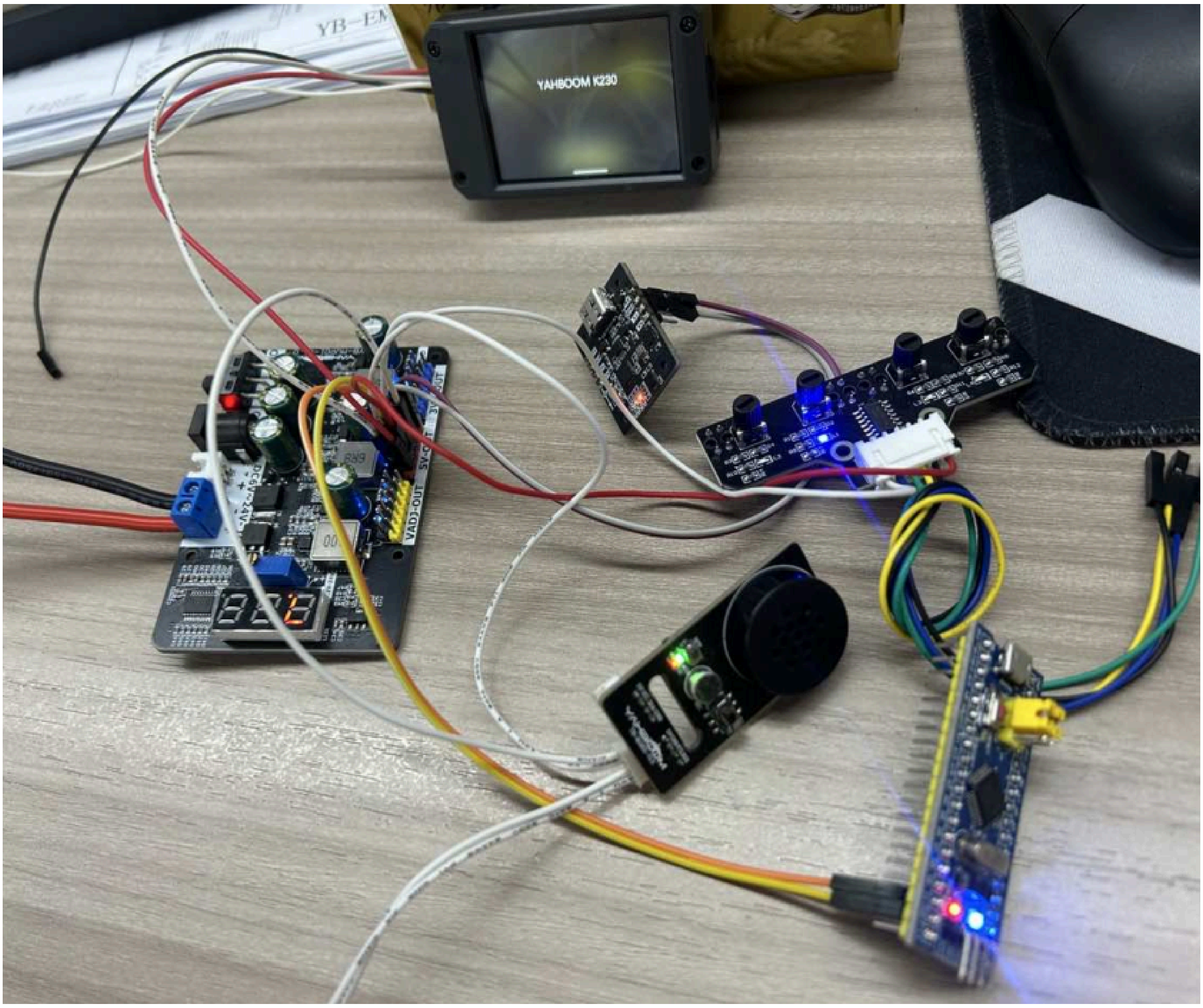
The densely populated area on the back of the module is covered with a large area of copper to enhance heat dissipation and support the attachment of heat sinks.




Large area thermal conductivity zone on the bottom

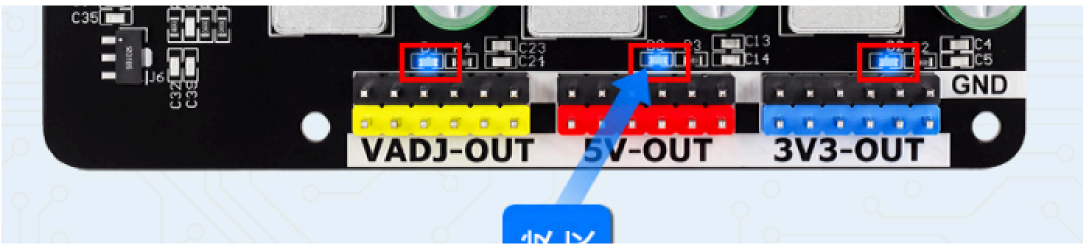
Support for attaching heat sinks
(*Need to prepare own heat sink by yourself)

When the device is operating under high-current output conditions, it is recommended to use **thick-gauge wires** to prevent the use of excessively thin wiring from compromising actual output performance. The stated maximum single-channel output of 5A represents the *aggregate* maximum current capacity for the entire device across all loads; if three output channels are enabled simultaneously, the current drawn from each individual channel must not exceed 4A. During operation, it is strictly forbidden to apply an external voltage in reverse polarity to the output terminals; this device's output terminals are designed solely for voltage output and must not be used as reverse-input ports. Users may connect multiple external modules to the device; for instance, connecting all external modules to the 5V output terminal—as illustrated below—allows for normal operation. Before powering up the system with multiple external modules connected, please meticulously verify the positive and negative polarity of all wiring; reversing the polarity is strictly prohibited, as doing so may result in the module burning out or sustaining permanent damage.



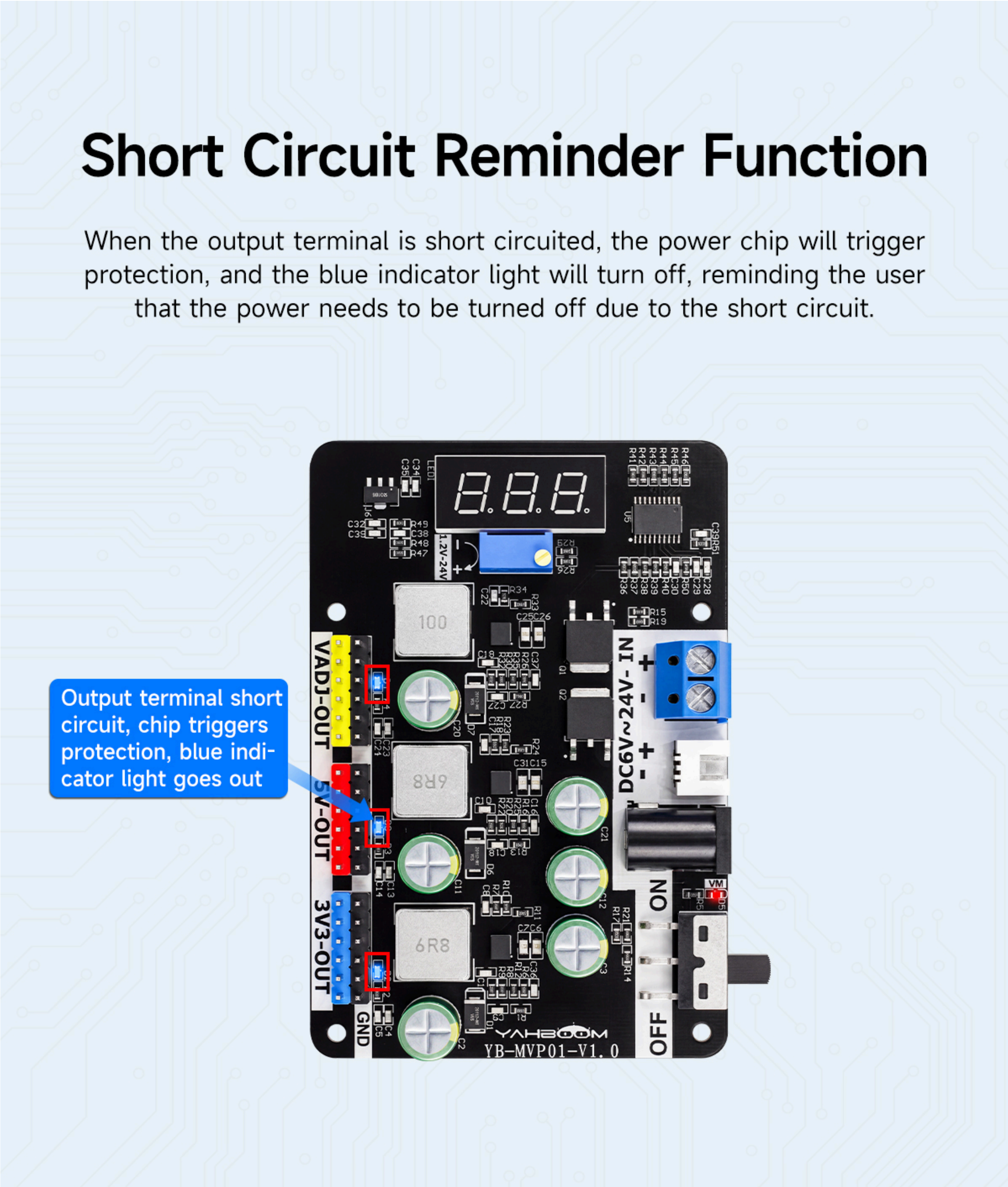

## 3. Indicator Light Functions

This module features a total of four LED indicator lights and one digital display tube. Located next to the power switch is a power indicator LED, which serves to confirm whether the power supply is properly connected.



The three LEDs shown here serve as status indicators for the DC power supply chip. When the chip is operating normally, the blue LED remains continuously lit. If the output load current exceeds 5A, the chip triggers its "hiccup mode" protection mechanism; the blue LED will then enter a "breathing" (pulsating) state, signaling that the current draw is excessive and the heat dissipation capacity is insufficient. In this scenario, you must reduce the load current or decrease the number of connected loads to ensure the board maintains proper thermal regulation.

If a short circuit occurs at the output terminals, the blue LED will turn off. In this event, immediately disconnect the power supply and troubleshoot the system to check for wiring errors or to verify that the underside of the circuit board has not come into contact with any metal objects.



**Digital Tube Display:**

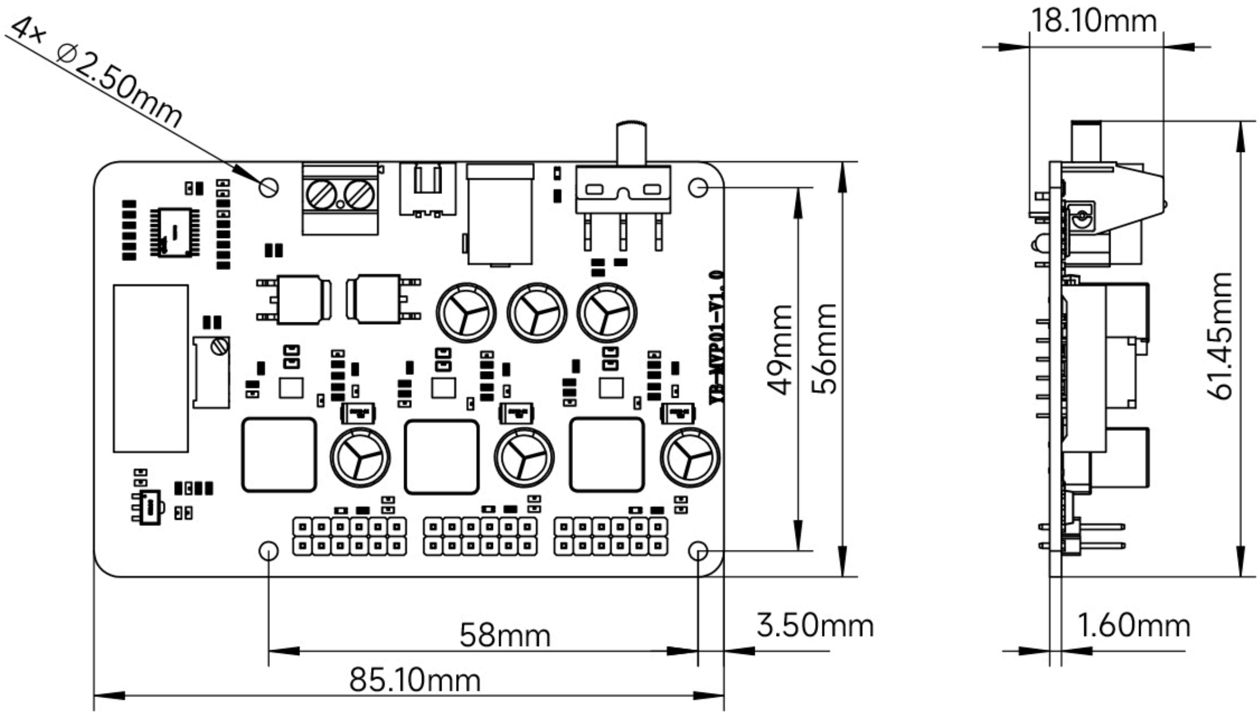
The digital tube display is controlled by a microcontroller. The necessary program has already been pre-loaded at the factory, so users can use the device immediately without any further configuration. When the adjustable output voltage is within the range of 1.2V to 24.5V, the digital tube displays the specific voltage normally. When the output voltage is between 24.5V and 26V, the digital tube will still display the specific voltage value, but it will begin to flicker. If the output voltage exceeds 26V, the digital tube will simply display "888," and the user must not apply an input power source exceeding 26V. Under normal operating conditions, it is recommended that users keep the input voltage within 24V. However, anticipating that some users might apply an input voltage exceeding 24V, the board has been designed with a voltage tolerance allowing for a maximum input of 26V. Applying an input voltage exceeding 26V will result in permanent damage to the board; therefore, users are strongly advised to operate within the 24V limit!

**Note:**


These are the programming pins for the microcontroller. The factory firmware has already been fully loaded; therefore, users must not attempt to reprogram the device on their own. The factory firmware is not provided with this module; if reprogramming becomes necessary, it will require a paid repair service.

## 4. Product Dimensions

# Dimensional Parameters




| Adjustable Step-down Voltage Regulator Module | | | |
|---|---|---|---|
| Voltage input | DC 6V~24V，＞1A | Voltage output | DC 3.3V、DC 5V、DC 1.2V~24V |
| Input interface | KF301-2P*1, XH2.54-2pin, DC5.5*2.5 | Output interface | 2.54-6p straight needle |
| Through-hole diameter | M2.5mm | Voltage meter accuracy | ±1% |
| Output current | ＜5A | DC voltage regulator chip | MP2491C*3 |
| Digital tube | Three anode digital tubes (highlighted in red) | Power indicator light | The power input indicator light is red and always on during normal input |
| Short-circuit indicator light | Output short circuit reminder indicator light. When the output terminal is short circuited, the blue indicator light goes out, reminding the user to turn off the power due to the short circuit | Protection circuit | Input anti-reverse protection, output over-current protection, hiccup current-limiting protection, over-voltage protection, short-circuit protection, over temperature shutdown protection |
| Product weight | 41.5g | Product dimensions | 85.1*56*18.1mm |
| Target audience | Various power supply competitions such as intelligent car competitions, electronic competitions, robot competitions, laboratory experiments, etc | | |

## 5. Important Precautions

- **Prohibition:** Do not connect the input voltage source to the output terminals.
- **Prohibition:** Do not apply an input voltage exceeding 24V to the input terminals.
- Ensure that the mounting screws are securely tightened to prevent them from falling out or causing the copper standoffs to shift, either of which could lead to a short circuit.
- During operation, place the product on a non-conductive flat surface to avoid accidental contact with metal objects, which could result in a short circuit.